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ASSOCIATION, SYNONYMITY, AND DIRECTIONALITY IN FALSE RECOGNITION¹

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When asked to indicate whether each of 200 orally presented words had appeared before (+) or not (—), students gave more plusses to common associates and synonyms of preceding words than to control words. In a 2nd experiment, false recognition errors were obtained when the preceding words associatively elicited the test words and when the associative relation was *bidirectional* but not when only the test words elicited the preceding words. The results of the 2nd experiment were taken as an indication that initial coding of words contributes to false recognition and that the phenomenon is not merely an artifact of testing for it. The occurrence of false recognition errors was taken as support for a characterization of words as complexes of attributes or features.

Even the bulkiest dictionaries present vocabulary as a list of unrelated items. Linguists, however, have made it clear that although the lexical component of language manifests less systematicity than syntax or phonology, it nevertheless has much more structure than is suggested by the arrangement of words in dictionaries (e.g., Chomsky, 1965, especially Ch. 4.; de Saussure, 1959; Katz & Fodor, 1963; Weinreich, 1964). Within the generative approach to language (Chomsky, 1965) a theory of semantics has recently been developing which has as its central notion the cross classification of words

into syntactically relevant categories, such as animate-inanimate, and human-nonhuman.

Psychologists too have recognized that words are organized and they have concerned themselves with the analysis of the psychological processes underlying this organization and with its consequences in verbal behavior. Experiments on semantic generalization (Feather, 1965), clustering in free recall (e.g., Cofer, 1965), not to mention work on free and controlled associations (e.g., Deese, 1965), can all be viewed as reflecting this concern. In these areas of psychological investigation, there is a general tendency, with some notable exceptions (e.g., Deese, 1965), to explain findings by reference to associative bonds between words. The

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present study was designed to challenge this approach.

In particular, our interest was aroused by a study of Underwood (1965) in which he used a method of continuous recognition originally introduced by Shepard and Teghtsoonian (1961). Underwood had Ss indicate for each of 200 words, presented by a tape recorder, whether it had occurred earlier in the list or not. He found that common associates of words which appeared earlier in the list were falsely recognized more often than control words.

The purpose of the first experiment to be reported below was to investigate whether relations of synonymy have similar effects to those of association. The constant use of paraphrasing in everyday life communication suggests that in coding for memory under normal conditions speakers retain primarily the semantic content of a message. Since synonyms have a large area of meaning in common, they would seem natural candidates for confusion in the kind of task used by Underwood. Of course, this experimental situation, unlike everyday life, puts a premium on verbatim coding, but it was reasoned that since associates proved to intrude in this situation synonyms certainly should.

The second experiment was suggested by Razran's study (1949) on semantic generalizations of conditioned responses. He found greater generalization when the test word strongly evoked, in a free association task, the conditioned word than when the conditioned word strongly evoked the test word. For instance, when a salivary response was conditioned to the word *dog* there was less generalization to the superordinate *animal* than to the subordinate *terrier*. In a free association task, subordinates tend to evoke superordinates more strongly than

superordinates evoke subordinates. His finding led Razran to the conclusion that semantic generalization was not an automatic process taking place during original conditioning but rather was an artifact of subsequent testing. Using unidirectional and bidirectional associates the present authors attempted to see whether false recognition was due to processes involved in initial coding of the stimulus or to confusion resulting from the presentation of its associate.

METHOD

Subjects

There were 28 male university students in the first experiment and 34 female students in the second.

Materials and procedure

Each of the two experiments contained 200 words which were recorded on magnetic tape at 10-sec. intervals. Each word was recorded twice in immediate succession to make sure that it would be heard. The tape was played to groups of five or six Ss. For each word, S had to indicate by a + or - whether it was "old" or "new." The Ss were instructed to guess when in doubt. The lists were constructed to contain three major categories of words: Preceding (P) words, Experimental (E) words, and Control (C) words. For each P word the list contained one or two E words, and for each E word one C word.

Experiment I.—The E words in the first experiment related to the P words in two ways: 24 as common associative responses (As) and 25 as synonyms (Ss). Table 1 presents the list of words used in this experiment. The words included as As were given as responses to their respective stimuli in a free association test by 27-70% of the 500 male college students Palermo and Jenkins (1964) used for their association norms. The words used as Ss were given as synonyms by 30-86% of the 50 students Jenkins and Palermo (1965) used for their synonym norms. As can be seen in Table 1, the E words were of four kinds: 15 As and 15 Ss had common P words (e.g., P: chair, A: table, S: seat), 4 As were related to Ps which had no obvious synonym (e.g., P: bed, A: sleep), 5 Ss

TABLE 1
WORDS USED IN EXP. I, THEIR ORDINAL POSITIONS AND NUMBER
OF FALSE RECOGNITION ERRORS

P		A			CA		S			CS	
Word	Position ^a	Word	Position ^b	Error	Word	Error	Word	Position ^b	Error	Word	Error
always	40	never	74	1	beside	1	forever	64	1	carefully	0
black	142	white	185	4	soft	3	dark	169	4	cold	1
girl	119	boy	167	6	air	5	female	157	1	marble	1
chair	60	table	88	7	valley	2	seat	78	0	job	2
high	132	low	173	1	ill	1	tall	163	2	safe	1
king	141	queen	181	4	dance	0	ruler	191	0	statue	0
live	32	die	52	1	see	3	exist	68	2	inform	0
needle	129	thread	153	2	nest	3	pin	175	1	jar	1
scissors	89	cut	110	0	look	0	shears	128	1	brine	0
thirsty	44	water	86	3	paper	0	dry	67	0	nice	0
whiskey	95	drink	135	7	spend	2	alcohol	115	1	corridor	0
bath	158	clean	186	5	small	3	shower	198	3	humor	1
now	145	then	176	2	near	9	immediately	192	2	certainly	4
over	36	under	61	1	within	0	above	83	4	early	2
tell	79	me	107	0	it	0	relate	123	2	deceive	0
Error Sub-totals				44		32			24		13
anger	166	mad	199	1	rough	3	mad	199	1	rough	3
carpet	159	rug	179	2	shed	2	rug	179	2	shed	2
eagle	111	bird	151	5	guard	3	bird	151	5	guard	3
kitten	54	cat	100	2	wool	0	cat	100	2	wool	0
swift	39	fast	58	0	third	0	fast	58	0	third	0
Error Sub-totals				10		8			10		8
bed	114	sleep	149	2	guess	0					
green	47	grass	91	2	heart	0					
his	97	hers	136	2	grim	1					
stem	35	flower	55	0	journal	0					
Error Sub-totals				6		1					
baby	134						infant	160	3	drama	1
citizen	37						member	57	2	winter	1
have	144						own	193	4	go	0
jump	49						hop	75	0	fetch	0
make	162						create	182	3	reduce	2
Error Sub-totals									12		4
Grand Totals				60		41			46		25

^a The positions indicated are for the third token of each P word.

^b The positions indicated are for the first order.

were related to Ps which had no common associates (e.g., P: baby, S: infant), and in five cases the same E word was both an A and an S (e.g., P: carpet, A, S: rug).

Our procedure for selecting control words differed from that employed by Underwood. Underwood's control words were common associates whose stimuli did not appear in the list. For instance, he used *down* as a control word because it is commonly given as a response to *up*, but *up* was not included in the list. This method does not provide a specific control word for each experimental word. Also, it implies that the only variable conceivably relevant to false recognition is associative connection—a stronger hypothesis than Underwood's experiment was designed to test. The large number of errors to the control words in Underwood's study suggests that factors other than associative

relation influence false recognition. Because of these considerations, our C words were selected so that each one of them would be similar to its experimental counterpart with respect to part of speech, frequency of usage as reflected in the Thorndike and Lorge (1944) G count, and number of syllables. In addition, the C words had to have no obvious relation to other words in the list.

The list thus contained 29 P words, 44 E words, and 44 C words. Each P word appeared three times in different positions before its E counterpart was heard. The three tokens of each P word were within 10–20 positions of each other, and the E word appeared within 18–50 positions after the last token of its corresponding P word. In the 15 cases where a single P word had two E counterparts, one A and one S, the two E words were located within 10–22 posi-

TABLE 2
WORDS USED IN EXP. 2, THEIR ORDINAL POSITIONS AND NUMBER
OF FALSE RECOGNITION ERRORS

P	Position*	E	Position	Errors	C	Position	Errors
P ← → E							
List 1							
black	120	white	169	1	fine	168	2
boy	70	girl	99	1	bird	101	0
chair	177	table	199	5	duty	198	3
king	82	queen	116	0	field	118	2
health	127	sickness	147	0	aspect	148	1
butter	79	bread	114	3	fault	115	2
high	131	low	164	5	all	162	4
square	102	round	142	4	young	144	3
green	104	grass	124	6	town	122	0
cold	159	hot	188	2	full	190	2
List 2							
white	120	black	169	1	fresh	168	1
girl	70	boy	99	0	car	101	0
table	177	chair	199	6	page	198	1
queen	82	king	116	3	star	118	1
sickness	127	health	147	0	church	148	1
bread	79	butter	114	3	evening	115	1
low	131	high	164	4	each	162	1
round	102	square	142	0	real	144	1
grass	104	green	124	1	large	122	1
hot	159	cold	188	2	just	190	1
Error							
Subtotals				47			28
P ← → E							
List 1							
bitter	125	sweet	155	2	best	153	0
fingers	78	hand	108	1	fact	107	1
how	156	now	180	5	why	179	2
long	130	short	161	1	rich	160	2
swift	149	fast	178	5	kind	176	6
bloom	154	flower	187	7	answer	189	4
sky	133	blue	166	1	free	167	0
whiskey	87	drink	121	1	break	119	1
tell	62	me	90	0	it	91	0
door	96	window	135	0	building	136	5
List 2							
appear	98	see	138	1	add	139	1
cottage	163	house	185	8	time	183	1
dream	75	sleep	111	4	price	113	0
heavy	69	light	93	2	stone	92	0
lift	72	carry	97	0	follow	95	0
loud	128	soft	157	4	glad	158	1
therefore	152	because	181	2	either	182	1
stomach	151	food	184	2	law	186	1
infant	143	baby	174	3	army	173	1
scissors	105	cut	146	1	ask	145	3
Error							
Subtotals				50			30
P ← → E							
List 1							
see	98	appear	138	1	begin	139	0
house	163	cottage	185	0	feather	183	3
sleep	75	dream	111	4	grant	113	0
light	69	heavy	93	2	double	92	1
carry	72	lift	97	0	drop	95	0
soft	128	loud	157	2	fair	158	2
because	152	therefore	181	0	without	182	0
food	151	stomach	184	0	lecture	186	0
baby	143	infant	174	1	jacket	173	0
cut	105	scissors	146	0	parchment	145	0
List 2							
sweet	125	bitter	155	1	active	153	1
hand	78	fingers	108	0	corners	107	0
now	156	how	180	3	yet	179	2
short	130	long	161	6	late	160	3
fast	149	swift	178	0	cruel	176	1
flower	154	bloom	187	0	crash	189	0
blue	133	sky	166	3	art	167	1
drink	87	whiskey	121	0	antique	119	0
me	62	tell	90	0	play	91	3
window	96	door	135	0	book	136	0
Error							
Subtotals				23			17
Error Grand							
Totals				120			75

* The positions indicated are for the second token of each P word.

tions of each other. Each C word was within two positions of its E word, with half the Cs preceding and half following the E words. The Cs for As will be referred to as CAs and those for Ss as CSs. The positions of P, E, and C words are indicated in Table 1.

In addition to words in these three categories, the list also contained 22 filler words. Of these, 20 appeared once each, one—twice, and one—three times. The fillers had no apparent relation to any of the other words in the list. They occupied positions 1–10 and other positions mostly in the first quarter of the list. The list was recorded in two orders. For the first order, the words were randomly arranged, within the constraints outlined above. The second order was identical to the first, except that the positions occupied by Ss and CSs were replaced by their respective As and CAs, and vice versa. This interchange was, of course, possible only for the words which had common Ps.

Experiment II.—In the second experiment there were 30 E words and all of them had an associative relation to the P words. The words used in this experiment and their positions can be seen in Table 2. The As were drawn from two sources. Some were given as responses to their respective stimuli by 22–73% of the 500 female college students in the Palermo and Jenkins (1964) norms and others by 26–74% of the 1,349 airmen in the Bilodeau and Howell (1965) norms. The 30 E words fell into three categories in terms of their relation to P words. In 10 P-E pairs, the E words were common responses to the P words as stimuli but they did not commonly elicit the P words as responses (e.g., bitter $\rightarrow$ sweet, P $\rightarrow$ E). In 10 other cases, the E words commonly elicited the P words as responses but this relation was not reciprocated by the P words (light $\leftarrow$ heavy, P $\leftarrow$ E). Finally, in 10 cases, the associative link went in both directions (king $\leftrightarrow$ queen, P $\leftrightarrow$ E). Most of the words considered as nonassociates were given by less than 10% of the respondents in the norms used and only one word reached 22% response commonality. For instance, while *sweet* was given by 53% of the Palermo-Jenkins Ss in response to *bitter*, only 2% gave *bitter* as a response to *sweet*. The words (List 1) that served as Ps for about half (16) of the Ss served as Es for the rest of the Ss (18, List 2), and vice versa. The mean association values for the directions indi-

cated are: P $\leftrightarrow$ E = 53%, P $\rightarrow$ E = P $\leftarrow$ E = 42%. In Exp. II, as in Exp. I, the characteristic that distinguished a P word from an E word was its ordinal position in the list: the P word came before its corresponding E word. Because each E word had a specific C word, the Cs for the two lists had to be different. In Exp. II, each P word appeared only twice. Thus, the distribution of words in the list was as follows: 60 positions were occupied by the Ps, 30 by the Es, and 30 by the Cs. The rest, 80 positions, were held by filler words. Thirty-five fillers appeared once each, 19—twice each, one—three times, and one—four times. The first 49 positions were taken by fillers, the rest of the fillers were scattered throughout. The second token of each P word was within 10–20 positions of the first token, and each E word was placed 20–50 positions following the second token of its P counterpart. In other respects the design of the second experiment was similar to that of the first.

RESULTS

Two types of errors were possible in these experiments: (a) identification of a new item as an old one (positive errors, or false recognition errors), and (b) identification of an old item as a new one (negative errors). Negative errors could have been made to the repetitions of the P words and filler words. False recognition errors could have been made to all other words. The number of errors of any kind was minimal. In Exp. II, it amounted to 6% out of the possible number of errors, and in Exp. I the number of errors reached only 5%.

Our interest in this paper is focused on the comparison of the number of errors made to the E words vs. the number made to the C words. The errors made in these two categories of words can be seen in Tables 1 and 2 for Exp. I and II, respectively. Inspection of the distribution of errors in Exp. I showed an atypically large number of errors to the P word “bath” (12 errors, while the next highest figure for a P word was 6) and to the CA

word "near" (9, next highest in this category being 5). For Exp. I, therefore, the comparisons between E and C words excluded the two rows (10 words) in which these two words appeared. After this exclusion, both the association and synonymy variables produced significant results, for the S-CS difference, $t(27) = 2.94$, $p < .01$, and for the A-CA difference, $t(27) = 2.52$, $p < .02$. Comparisons for the complete data without exclusions yielded significant results for S-CS, $t(27) = 2.42$, $p < .05$, but not for A-CA, $t(27) = 1.85$, $p < .10$.

The exclusions thus do not alter the status of synonymy as a relevant variable, they affect seriously only the association variable. Since the main object of this study was to weaken the omnipotence of associative relations in verbal behavior, the exclusion procedure seems theoretically conservative. As will be seen below, another type of experimental error, "misperception," militated in the present study against the association variable.

The means for the 23 entries involved in the S-CS comparison are: $S = 1.46$, $CS = .71$, and the means for the 22 A-CA entries are: $A = 1.89$, $CA = 1.03$. The five E words which were related to the P words as both associates and synonyms, and their corresponding C words, were included in both the A-CA and the S-CS comparisons, but as can be seen in Table 1 these words contributed very little to the differences obtained.

The higher error rate for the A and CA categories as compared to the S and CS categories may be due to the higher frequency of these words. While only 5 A and 5 CA words had Thorndike-Lorge frequencies under AA, 16 S words and 15 CS words fell below this level. The more frequent the words, the greater the likelihood that they will have interrelationships amongst them-

selves. Such relationships, beyond those under experimental control, will tend to contribute to false recognition. Generally, it is practically impossible to construct a list of 200 words which will not contain relations other than those built-in experimentally. The best one can achieve is the elimination of any glaring nonexperimental relations and then depend on chance to distribute the remaining relations equally among all conditions.

In Exp. II, the statistical analysis for the pooled data of Lists 1 and 2 reveals that significantly more false recognition errors were made to the E words than to the C words when the E words were bidirectionally associated with P words, $P \leftrightarrow E$; means: $E = 1.38$, $C = .82$, $t(33) = 2.81$, $p < .01$, and when the associative link was in the forward direction, $P \rightarrow E$; $E = 1.47$, $C = .88$, $t(33) = 2.23$, $p < .05$, but not when the association was backward, $P \leftarrow E$; $E = .68$, $C = .50$, $t(33) = .65$, *ns*. In separate analyses for List 1 and List 2, the pattern of results for List 2 was identical to the pattern for the pooled data but none of the E-C comparisons for List 1 reached significance.

Why was the pattern of results in List 2 not duplicated in List 1? It may be seen by inspecting Table 2 that the number of errors to the E words in List 1 was roughly the same as in List 2. The difference between the two lists is that the number of errors to the C words in List 1 was more than double that in List 2 for the $P \leftrightarrow E$ and $P \rightarrow E$ categories. It is possible to blame the higher error score of the C words in List 1 on gratuitous relations between some of these words and other words in the list. For instance, *kind* which was used as a control for the adjective *fast* contributed a very high number of errors (6). This control could have been perceived by Ss as a synonym of the noun *type* which ap-

peared earlier in the list as a filler. Although such ad hoc explanations are of little value, because one can probably find some nonexperimental relation for most of the words in the list, the point stands that uncontrolled (and uncontrollable) relations, whatever they are, played a role in this experiment, and could account for the disproportionate number of errors in the C category of List 1.

After completion of the experiments the suspicion arose that some words may not have been heard as the words they had been intended to be. To check on this possibility, different Ss were asked to listen to the tapes and to write down what they heard. Eight Ss did this for the tape used in Exp. I and four for the tape used in Exp. II. This test revealed some misperception, mostly due to homophones. The misperceptions of one or more Ss were distributed among the word categories as follows: Exp. I, two P words, four A words, six CA words, one S word, and one CS word; Exp. II, seven P and E words, and four C words.

It is apparent that in Exp. I there was a greater concentration of misperceptions in the A category words than in the other categories. Because of this it is likely that the *t* value for the difference between A and C words underestimates the true contribution of associative relations to false recognition. However, this problem is not too serious for the present study because its purpose was not to ascertain the exact weight of associations in determining false recognition but rather to introduce synonymity as an additional variable.

DISCUSSION

In interpreting the results of the second experiment it must be kept in mind that the forward ($P \rightarrow E$) and backward ($P \leftarrow E$) conditions differed not only in temporal order of the free association

stimuli and responses relative to each other but also in the number of times the associative stimuli appeared. In the forward condition where the associative stimuli served as P items, they appeared twice, but in the backward condition, where they served as E items, they appeared only once. For this reason it is not possible to conclude that backward associative relations cannot produce false recognition. But one can conclude that forward relations do result in such an effect, thus establishing that initial coding plays a role in false recognition, and that it is not merely an artifact of testing for it. The discrepancy between this conclusion and Razran's is not too disturbing, because of the basic methodological flaws in his experiment recently summarized by Feather (1965). In fact, despite the gadgetry involved in semantic generalization experiments—perhaps, because of it—it seems that the judgmental procedure employed in the present experiments taps the underlying semantic processes more directly than the semantic generalization procedure. In semantic generalization, the CR transfers, according to this view, to words judged, mistakenly in the case of the test words, by Ss (not necessarily in full awareness) as having been previously conditioned. The generalization gradient, whenever it obtains, may reflect the degree of confidence associated with such judgments. Such mistaken judgments reflect underlying processes of word coding.

What are these processes? A model is needed that would account both for Ss' pronounced ability to recognize correctly new and old words and for the systematic false recognition errors found. It is clear that S must somehow mark off the words on the list from the tens of thousands of words in his vocabulary. This marking off may be achieved in the following way. Assume that, when activated, the neural processes giving a word its identity leave a trace which is "dark" and "heavy" at first and with the passage of time wears out and becomes fainter and fainter. A mechanism of this sort would keep the time of words and enable S to judge each word as to whether it was heard in the experimental session or prior to it. On

the basis of this information *S* could classify a word as new or old.

This timing notion can thus account for *Ss'* correct identification of words. But it cannot explain why *Ss* made the systematic false recognition errors described. In order to account for these errors it must be assumed that the word is not the ultimate unit of coding. If it were, associates and synonyms should not produce more errors than control words. Rather, our finding, along with the common phenomenon of paraphrasing and related observations (e.g., Broadbent, 1964; Brown & McNeill, 1966; Yavuz & Bousfield, 1959), supports the conception of words as complexes of features. According to this view, each word consists of a set of features or attributes which uniquely characterize it and distinguish it from all other words in the vocabulary system. The features are of many different kinds and involve semantic, syntactic, phonological, and for literates, orthographic aspects. For instance, the feature characterization of *table* would describe it as "a piece of furniture," "a noun," "having a [t] sound in initial position," "an (e) letter in final position," etc. On this conception, the encoding of a word would correspond to a *simultaneous* activation of a set of features. Many of the semantic features and all of the features in the other three categories would be common to large segments of *S's* lexicon, but some semantic features would be specific to a single item or to a small group of items (see Katz & Fodor, 1963). The general features serve to relate various lexical items to each other. For instance, many English words begin with a (t), and all these words constitute, in some sense, a category whose members share this feature. But the fact that a *table* stands at a certain height, broadly defined, in relation to its other dimensions could be thought of as an idiosyncratic feature.

Within the framework of the feature analysis, different responses to the same word share to a great extent the same processes. When an *S* is asked in a free-association test to give the first word that comes to his mind, he focuses on some of

the features of the word, perhaps the ones which appear to him most salient at the time. When he is asked to give a synonym, he focuses again on a subset of the feature complex. In this case the features relevant for the determination of the response are semantic. When he is asked to produce rhymes, he focuses on phonological features, and when asked to respond with words of the same letter length as the stimuli, he focuses on orthographic features. According to this view, associative responses and so-called categorical responses do not entail totally different processes, they result mainly from different selections of features.

The idea of feature coding can account for the false recognition errors resulting from associative and synonymy relations. When a new word is heard which shares some significant features with an old word, *S* may be led mistakenly to "disregard" the distinguishing features and consider the two as identical. The assumption implied here is that not all features carry equal weight. When a word is heard, some of the features potentially associated with it may not get activated and even if activated not all features leave noticeable traces.

In conclusion, we would like to restate the hypothesis that words are not stored as words but as complexes of features. When words are used they are not reproduced from memory but rather reconstructed from their component features.

REFERENCES

- BILODEAU, E. A., & HOWELL, D. C. Free association norms by discrete and continued methods. Technical Report No. 1, 1965, Tulane University, Contract Nonr. 475 (10), Office of Naval Research, New Orleans.
- BROADBENT, D. E. Perceptual and response factors in the organization of speech. In A. V. S. de Reuck & Maeve O'Connor (Eds.), *Disorders of language; Ciba Foundation Symposium*. Boston: Little, Brown, 1964.
- BROWN, R., & MCNEILL, D. The "tip of the tongue" phenomenon. *J. verbal Learn. verbal Behav.*, 1966, 5, 325-337.
- CHOMSKY, N. *Aspects of the theory of syntax*. Cambridge: M.I.T. Press, 1965.

- COFER, C. N. On some factors in the organizational characteristics of free recall. *Amer. Psychologist*, 1965, 20, 261-272.
- DEESE, J. *The structure of associations in language and thought*. Baltimore: Johns Hopkins Press, 1965.
- DE SAUSSURE, F. *Course in general linguistics*. (Trans. by W. Baskin) New York: Philosophical Library, 1959.
- FEATHER, B. W. Semantic generalization of classically conditioned responses: A review. *Psychol. Bull.*, 1965, 63, 425-441.
- JENKINS, J. J., & PALERMO, D. S. *Synonym responses to the stimulus words of the Palermo-Jenkins word association list*. (Res. Bull. No. 55). University Park: Department of Psychology, Pennsylvania State University, 1965.
- KATZ, J. J., & FODOR, J. A. The structure of a semantic theory. *Language*, 1963, 39, 170-210.
- PALERMO, D. S., & JENKINS, J. J. *Word association norms; grade school through college*. Minneapolis: University of Minnesota Press, 1964.
- RAZRAN, G. Semantic and phonetographic generalizations of salivary conditioning to verbal stimuli. *J. exp. Psychol.*, 1949, 39, 642-652.
- SHEPARD, R. N., & TEGHTSOONIAN, M. Retention of information under conditions approaching a steady state. *J. exp. Psychol.*, 1961, 62, 302-309.
- THORNDIKE, E. L., & LORGE, I. *The teacher's word book of 30,000 words*. New York: Teachers College, Columbia University, 1944.
- UNDERWOOD, B. J. False recognition by implicit verbal responses. *J. exp. Psychol.*, 1965, 70, 122-129.
- WEINREICH, U. Webster's third: A critique of its semantics. *Int. J. Amer. Ling.*, 1964, 30, 405-409.
- YAVUZ, H. S., & BOUSFIELD, W. A. Recall of connotative meaning. *Psychol. Rep.*, 1959, 5, 319-320.

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- Transfer-Activated Response Sets: The Effect of Overtraining and Percentage Items Shifted on a Verbal Discrimination Shift: Coleman Paul,* Marilyn Mereness, Charles Callahan, and Kenneth Wilhelm: Department of Psychology, Adelphi University, Garden City, Long Island, New York 11530.
- Effects of the Response-Shock Contingency on the Facilitation of Discrimination Performance: James M. McCroskery and John W. Donahoe*: Department of Psychology, University of Kentucky, Lexington, Kentucky 40506.
- Replicability of an Optimal Delay of Reinforcement Result in Instrumental Eyelid Conditioning: Louise E. Cerekwicki, Barry H. Kantowitz, and David A. Grant*: Department of Psychology, University of Wisconsin, Charter at Johnson, Madison, Wisconsin 53706.
- Serial Acquisition as a Function of Number of Successively Occurring List Items: James F. Voss*: Department of Psychology, University of Pittsburgh, Pittsburgh, Pennsylvania 15213.
- Searching Circular Sequences: Robert J. Weber,* Michael Cross, and Myrna Carlton: Department of Psychology, Oklahoma State University, Stillwater, Oklahoma 74074.
- Long-Term Probability Learning with a Random Schedule of Reinforcement: Morton Y. Friedman,* Edward C. Carterette, and Norman H. Anderson: The Psychological Laboratory, University of Cambridge, Downing Street, Cambridge, England.
- Response Characteristics and the Orienting Reflex: Joseph Germana*: Department of Psychology, Virginia Polytechnic Institute, Blacksburg, Virginia 24061

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